Methods: Predicate frequencies in SemMedDB-KGX and recall against CTD curated chemical–disease associations

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1 Overview

We performed two analyses on the Biolink-normalized Knowledge Graph Exchange (KGX) release of the Semantic MEDLINE Database (SemMedDB) [1, 2]: (i) the frequency of each (subject category, predicate, object category) triple across the edge set, and (ii) the recall of SemMedDB chemical–disease edges relative to the manually curated chemical–disease associations in the Comparative Toxicogenomics Database (CTD) [3], evaluated only on PubMed articles indexed by both resources. Both analyses were implemented as SQL queries in DuckDB [5] and orchestrated with Snakemake [6].

2 Data sources

SemMedDB-KGX. We used the "uncapped" SemMedDB-KGX archive `kg2.10.3_semmeddb_dogpark_uncappe` from the Translator-CATRAX SemMedDB-KGX project [2], and extracted two files from it:

- `normalized_nodes.jsonl`: one record per node, giving its CURIE (a UMLS CUI), an ordered list of Biolink categories [4], and the node normalizer's `equivalent_identifiers` cross-references;
- `normalized_edges.jsonl`: one record per (subject, predicate, object) edge, with Biolink predicates and the list of supporting PubMed identifiers (`publications`).

This edge set is a filtered subset of raw SemMedDB. Upstream it was filtered by minimum supporting-publication count, by Biolink domain/range constraints, and by removing predicates excluded by BioThings Explorer (see the SemMedDB-KGX documentation). So all frequencies and recall estimates below describe this curated edge set, not the raw SemMedDB `PREDICATION` table.

CTD. We downloaded CTD's curated chemical–disease associations file (`CTD_curated_chemicals_diseases.cs` from `https://ctdbase.org/reports/`. Each row gives a chemical (a MeSH accession), a disease (a MeSH or OMIM CURIE), a `DirectEvidence` value (`therapeutic` or `marker/mechanism`), and a pipe-delimited list of the PubMed IDs that support that evidence type. CTD does not version this file at the URL, so the release we used is the one current on the download date.

3 Category assignment

Each node was assigned its *most specific* Biolink category: the first element of its `category` list, following the KGX convention of ordering categories from most to least specific. The `biolink:`

prefix was removed from categories and predicates.

4 Predicate frequency estimation

We joined every edge to the most specific category of its subject and object nodes, and grouped the edges by (subject category, predicate, object category). Each group reports two counts:

- n_{edges}: the number of distinct edges in the group;
- $n_{\text{publications}}$: the sum, over those edges, of the length of each edge's `publications` list. This is a count of edge–article support instances, not of distinct articles.

Edges whose subject or object had no node record are excluded by the inner join. We also reported counts separately for a configured set of triples of interest (for example, `Procedure`–`has_input`–`PhysicalEntity` and `Disease`–`located_in`–`GrossAnatomicalStructure`). A triple that does not occur is reported with zero counts.

5 Recall against CTD

5.1 Mapping SemMedDB concepts to MeSH

SemMedDB identifies concepts by UMLS CUI, while CTD uses MeSH. We mapped CUIs to MeSH using the `MESH:` entries in each node's `equivalent_identifiers`. Because this mapping comes from the node normalizer's existing cross-references, it does not require a UMLS Metathesaurus (MRCONSO) license. A CUI with several MeSH cross-references keeps all of them, so the mapping can be many-to-many.

Chemical-like and disease-like nodes were defined by their most specific category:

- chemical: `SmallMolecule`, `ChemicalEntity`, `Drug`, `ChemicalMixture`;
- disease: `Disease`, `PhenotypicFeature`, `DiseaseOrPhenotypicFeature`.

5.2 SemMedDB chemical–disease assertions

We kept only edges with a chemical-like subject and a disease-like object, so edges asserted in the opposite direction (disease → chemical) are not considered. Each retained edge was expanded over its `publications` list, and its subject and object were mapped to MeSH. This gives the set of distinct tuples

$$S = \{(c, p, d, m)\},$$

where c is a chemical MeSH ID, p a Biolink predicate, d a disease MeSH ID and m a PubMed ID. Subjects or objects without a MeSH cross-reference drop out at this step.

5.3 CTD reference assertions

Each CTD row was expanded over its `DirectEvidence` values and its PubMed IDs, and the chemical accession was prefixed with `MESH:`. This gives the set of distinct tuples

$$C = \{(c, d, e, m)\},$$

where $e \in \{\texttt{therapeutic}, \texttt{marker/mechanism}\}$. Rows whose disease is an OMIM identifier (not MeSH) cannot match SemMedDB's MeSH-mapped diseases, so we excluded them and recorded how many there were.

5.4 Common-article anchor

A CTD assertion can only be recovered by SemMedDB if SemMedDB processed the same article, so we restricted the evaluation to articles both resources cover. We defined the common PubMed set as

$$M = \pi_m(C) \cap \pi_m(S),$$

where π_m is the set of distinct PubMed IDs in a table, and $\pi_m(S)$ is taken over chemical–disease edges only. The evaluation set is $C_M = \{(c, d, e, m) \in C : m \in M\}$, and SemMedDB is restricted the same way to S_M.

5.5 Matching criteria

Each CTD assertion $(c, d, e, m) \in C_M$ was scored against two criteria of increasing strictness:

Entity match: SemMedDB asserts *some* relation between c and d in article m: $\exists\, p : (c, p, d, m) \in S_M$.

Predicate match: SemMedDB asserts such a relation with a predicate consistent with CTD's evidence type: $\exists\, p \in P(e) : (c, p, d, m) \in S_M$.

The allowed predicate sets $P(e)$ are given in Table 1. We chose them conservatively: generic predicates that do not reflect curated evidence (`related_to`, `coexists_with`, `located_in`, `produces`, `has_part`, `derives_from`, `occurs_in`, `precedes`) count toward neither evidence type. For each CTD assertion we also recorded the full list of SemMedDB predicates that link the same chemical, disease and article, so that alternative predicate mappings can be assessed afterwards.

Table 1: SemMedDB predicates counted as consistent with each CTD `DirectEvidence` type.

CTD `DirectEvidence`	Allowed SemMedDB predicates $P(e)$
`therapeutic`	`treats_or_applied_or_studied_to_treat`
`marker/mechanism`	`causes`, `affects`, `disrupts`, `predisposes_to_condition`, `exacerbates_condition`, `interacts_with`

5.6 Recall

For each evidence type e, and for all types pooled, recall is the fraction of CTD assertions in the evaluation set that are matched:

$$R_e^{\text{entity}} = \frac{|\{x \in C_M^{(e)} : x \text{ entity-matched}\}|}{|C_M^{(e)}|}, \qquad R_e^{\text{predicate}} = \frac{|\{x \in C_M^{(e)} : x \text{ predicate-matched}\}|}{|C_M^{(e)}|},$$

where $C_M^{(e)}$ is the subset of C_M with evidence type e. The unit of evaluation is a (chemical, disease, evidence type, article) tuple. In the pooled estimate, a chemical–disease pair supported by the same article under both evidence types therefore counts twice. By construction, $R_e^{\text{predicate}} \le R_e^{\text{entity}}$.

5.7 Diagnostics

We also report: the number of common PubMed IDs and the number of distinct PubMed IDs on each side; the number of CTD rows and distinct chemical–disease pairs excluded for OMIM disease identifiers; and the fraction of chemical-subject and disease-object CUIs (among chemical–disease edges) that have a MeSH cross-reference.

6 Limitations

- **Filtered edge set.** SemMedDB-KGX excludes low-support and constraint-violating predications, so recall describes this curated release rather than raw SemMedDB, and is likely lower than raw SemMedDB's.
- **Identifier mapping.** CUIs without a MeSH cross-reference, and CTD diseases identified only by OMIM, cannot be matched. Matching is on exact MeSH identifiers, with no credit for parent or child concepts in the MeSH hierarchy.
- **Direction and category.** Only chemical → disease edges are considered, and only for nodes whose *most specific* category is in the configured sets. A chemical whose most specific category is outside these sets (for example, a protein-like category) is not considered.
- **Predicate mapping.** Predicate recall depends on the choice of $P(e)$ in Table 1. Entity recall gives an upper bound that does not depend on that choice.

7 Implementation and reproducibility

The analyses are implemented in a Snakemake (v9.27.0) workflow. The predicate-frequency and recall queries run in DuckDB 1.5.5 (Python 3.12), inside a pinned container image; the downloads run in a separate Debian-based image containing `curl` and `tar`. JSON Lines were read with an explicit column projection, which avoids loading the large embedded supporting-text field, and with a 64 MiB maximum record size, because individual heavily cited edges exceed DuckDB's 16 MiB default. DuckDB was given an explicit memory limit and an on-disk spill directory so that it could process the inputs out of core. The workflow ran on the UCSF CoreHPC Slurm cluster. Because the compute nodes have no outbound network access, both source datasets were downloaded on a login node beforehand. The workflow writes the following outputs: `predicate_frequencies.csv` and `predicate_frequency_lookup.tsv` for the frequency analysis, and `ctd_recall_pairs.csv` (one row per CTD assertion in C_M, with its match flags and the matched SemMedDB predicates) and `ctd_recall_summary.csv` (recall by evidence type and pooled) for the recall analysis.

References

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